
From EEG to Image: Decoding Visual Mental Imagery via Language-Mediated Generation and Direct CLIP Alignment

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 When a person closes their eyes and imagines an object, their brain generates elec-
2 trical activity that partially reflects the imagined object’s visual properties. If we
3 could decode these brain signals and render them as images, we would open a direct
4 channel from thought to visual output—a capability with profound implications
5 for assistive communication, brain-computer interfaces, and our scientific under-
6 standing of mental imagery. In this work, we address the problem of translating
7 raw electroencephalography (EEG) signals, recorded while subjects *imagine* visual
8 objects, into photorealistic images of those objects. We design and compare two
9 complete end-to-end pipelines evaluated on a 10-class cross-subject benchmark
10 using the Figshare Visual Imagery EEG dataset (22 subjects). Both pipelines
11 use a frozen CSBrain foundation model as a shared EEG feature extractor and
12 an anatomy-aware token reduction module. **Pipeline 2 (EEG-LLM)** routes EEG
13 features through a language model: a projection network maps EEG representations
14 into TinyLlama-1.1B’s embedding space, LoRA fine-tuning teaches the language
15 model to generate class-descriptive text from brain signals, and Stable Diffusion 2.1
16 synthesises the final image. **Pipeline 3 (EEG-CLIP)** eliminates the language
17 step entirely: a compact 4.8M-parameter EEGCLIPMapper—a Q-Former-style
18 transformer network—directly aligns EEG features with CLIP’s visual embedding
19 space and conditions Stable Diffusion v1.5. Both pipelines substantially exceed
20 the 10% chance baseline. Pipeline 3 achieves **35.8%** test accuracy versus 32.6%
21 for Pipeline 2, at a fraction of the computational cost (4.8M vs. $\sim$ 1.2B effective
22 parameters). A key finding is that CLIP *image* embeddings of stimulus photographs
23 (mean pairwise cosine similarity $\approx$ 0.54) are far more discriminative contrastive
24 training targets than CLIP *text* embeddings of class names ($\approx$ 0.90), explaining why
25 direct visual alignment outperforms language-mediated decoding.

26 1 Introduction

27 1.1 The Problem: Reading Visual Thought

28 When you imagine a dog, your brain does not merely retrieve a verbal label or an abstract concept—it
29 constructs an internal visual representation that activates many of the same neural circuits involved in
30 *actually seeing* a dog [?]. This process, called visual mental imagery, produces measurable electrical
31 activity in the brain that carries, at least partially, information about the imagined object’s category.
32 The ability to decode this activity and render it as a visible image would mean that a person could
33 communicate visual intent or mental content without any motor action—purely through thought.

34 This capability is not merely an academic curiosity. For individuals who have lost the ability to speak
35 or move—due to conditions such as amyotrophic lateral sclerosis (ALS), locked-in syndrome, or
36 severe cerebral palsy—brain-computer interfaces (BCIs) represent one of the few remaining channels
37 of communication with the outside world. Current BCIs largely focus on spelling text through
38 imagined motor commands or eye movements [?]. An imagery-to-image BCI, by contrast, could
39 allow such individuals to express visual concepts, communicate needs, or create—without relying
40 on any residual motor function. Understanding how mental imagery is represented in brain signals
41 is therefore both a fundamental scientific question and a practical engineering problem with direct
42 humanitarian relevance.

43 1.2 Why EEG?

44 Brain signals can be recorded by several modalities. Functional magnetic resonance imaging (fMRI)
45 measures changes in blood oxygenation and provides high spatial resolution (millimetre scale),
46 allowing researchers to identify which brain regions activate during a task. However, fMRI is
47 expensive (thousands of dollars per hour), requires the subject to lie motionless inside a large
48 magnetic scanner, has poor temporal resolution (1–2 seconds per measurement), and is completely
49 impractical for everyday BCI use.

50 Electroencephalography (EEG) works on a fundamentally different principle: metal electrodes placed
51 on the scalp measure the tiny electrical potentials generated by the synchronised firing of large
52 populations of cortical neurons. EEG offers millisecond temporal resolution, captures dynamic
53 temporal patterns in brain activity (alpha waves, gamma bursts, event-related potentials), is non-
54 invasive and safe, requires only a wearable headset, and is comparatively inexpensive. These
55 properties make EEG uniquely suited for practical BCI applications.

56 The trade-off is signal quality. EEG suffers from several sources of noise: muscle activity (especially
57 from jaw and eye movements), environmental electrical interference, volume conduction (signals from
58 a localised brain region spread and are detected by many electrodes simultaneously), and inherent
59 inter-subject variability in skull thickness and brain anatomy. As a result, EEG signals carrying
60 mental imagery information are considerably weaker and noisier than fMRI signals. Achieving
61 high-quality image reconstruction from EEG is therefore a substantially harder problem than from
62 fMRI.

63 1.3 The Core Challenge: Mental Imagery Under Cross-Subject Conditions

64 Most prior deep learning work on brain-to-image reconstruction has focused on *visual perception*—
65 reconstructing what a subject is currently looking at—rather than *visual imagery*—reconstructing
66 what a subject is currently imagining [Ozcelik and VanRullen, 2023, Scotti et al., 2023]. Perception
67 and imagery recruit overlapping but distinct neural circuits, and critically, perceptual EEG contains
68 much stronger stimulus-locked responses (signals that consistently peak at the same latency after
69 the stimulus onset). Imagery EEG lacks this strong temporal anchor, making it inherently harder to
70 decode.

71 A second, equally important challenge is *cross-subject generalisation*. EEG signals are highly
72 individual: two people imagining the same object produce qualitatively different spatial patterns
73 of activity due to anatomical differences, cognitive style differences, and varying levels of imagery
74 vividness. A model that learns subject-specific features during training will simply not work on a
75 new person. For any real-world BCI application, the system must be able to generalise to users it has
76 never seen before—a property we call *zero-shot cross-subject generalisation*. This is the setting we
77 evaluate in this paper.

78 1.4 What Has Been Tried Before

79 Early approaches to EEG-based image generation used class-conditional generative adversarial
80 networks (GANs). ThoughtViz [Tirupattur et al., 2018] and Brain2Image [Kavassidis et al., 2017]
81 trained GAN generators conditioned on EEG class embeddings, producing blurry but semantically
82 relevant images across a handful of categories. These methods used subject-dependent training and
83 were not evaluated cross-subject. More importantly, GAN-based generation at that time could not
84 produce photorealistic images. The advent of large-scale generative models changed this. Stable

Diffusion [Rombach et al., 2022], a latent diffusion model trained on billions of image-text pairs, can produce photorealistic images from text or CLIP [Radford et al., 2021] embedding conditioning. Bai et al. [Bai et al., 2023] combined EEG encoding with CLIP-guided Stable Diffusion and demonstrated that CLIP alignment substantially improves both classification accuracy and image quality—providing the most direct precedent for our work.

On the fMRI side, Ozcelik & VanRullen [Ozcelik and VanRullen, 2023] and MindEye [Scotti et al., 2023] showed that mapping fMRI features to CLIP space and conditioning a diffusion model can reconstruct perceived images at impressive quality. While these methods are not directly applicable to EEG (different modality, different SNR regime, different spatial resolution), they establish the general principle that aligning brain signals to CLIP space is a viable and powerful strategy.

A parallel line of work has developed self-supervised *foundation models* for EEG—models pre-trained on large, diverse EEG corpora that learn transferable representations applicable to many downstream tasks [Kostas et al., 2022, Jiang et al., 2024, Zhu et al., 2024]. These models, analogous to BERT or GPT in natural language processing, provide a strong starting point for downstream EEG decoding without requiring task-specific feature engineering.

The gap. Despite this progress, two questions remain unanswered: (1) Is it better to route EEG features through language (EEG $\rightarrow$ text $\rightarrow$ image) or to align directly to visual embedding space (EEG $\rightarrow$ CLIP $\rightarrow$ image) for mental imagery decoding? (2) When training the CLIP alignment, should we use CLIP image embeddings of the stimulus photographs or CLIP text embeddings of class names as training targets? These questions have not been systematically studied for visual imagery EEG under cross-subject conditions.

1.5 Our Approach and Findings

We design, train, and compare two end-to-end EEG-to-image pipelines that answer these questions head-on.

Pipeline 2 (EEG-LLM)¹ routes through language. EEG features are projected into a large language model’s embedding space, the model (TinyLlama-1.1B, fine-tuned with LoRA) generates a class-descriptive text prompt, and Stable Diffusion 2.1 synthesises the final image from that text.

Pipeline 3 (EEG-CLIP) bypasses language entirely. We introduce the **EEGCLIPMapper**, a compact Q-Former-inspired transformer network that maps EEG features directly to CLIP’s visual embedding space. The predicted CLIP embedding directly conditions Stable Diffusion v1.5. We train the mapper with a carefully designed composite loss that simultaneously promotes class discriminability, contrastive alignment with CLIP image prototypes, and sequence-level alignment with CLIP text representations.

Our key findings are: (1) Pipeline 3 achieves 35.8% test accuracy versus 32.6% for Pipeline 2—direct CLIP alignment outperforms language mediation. (2) The EEGCLIPMapper does this with 4.8M parameters versus ~ 1.2 B for Pipeline 2 and trains $\sim 10\times$ faster. (3) The reason direct alignment wins is quantifiable: CLIP image embeddings of the 10 stimulus classes have mean pairwise cosine similarity 0.54 (the classes are well-separated in this space), while CLIP text embeddings have mean pairwise similarity 0.90 (nearly indistinguishable). Text targets provide almost no useful training signal for contrastive learning.

Contributions:

1. Two complete, reproducible EEG-to-image pipelines on the Figshare 10-class cross-subject benchmark, including all preprocessing, training, and evaluation code.
2. The **EEGCLIPMapper**: a 4.8M-parameter Q-Former-style network that bridges 20 EEG region tokens to the 77-token, 768-dimensional CLIP conditioning format required by Stable Diffusion.

¹We label our systems Pipeline 2 and Pipeline 3 to maintain consistency with our broader experimental framework, in which Pipeline 1—a simple linear probe (logistic regression on mean-pooled CSBrain features) that achieved 18.3% test accuracy—served as an early cross-subject sanity check confirming that CSBrain features carry class-discriminative information. Pipeline 1 is not described in detail as it involves no novel components; we include it here only to clarify the numbering convention.

3. An empirical and geometric explanation of why CLIP image embeddings (pairwise sim ≈ 0.54) are far superior contrastive targets to CLIP text embeddings (≈ 0.90) for EEG alignment.
4. A two-phase training strategy that prevents conflicting gradient directions from the MSE and cosine alignment objectives, yielding 2–4% accuracy gains applicable to general multimodal CLIP-alignment problems.

2 Related Work

EEG decoding and foundation models. EEG decoding—classifying cognitive states or imagined stimuli from raw electrode recordings—has been studied for decades. Classical methods relied on hand-crafted spectral features such as event-related synchronisation/desynchronisation and Common Spatial Patterns (CSP) [Blankertz et al., 2007], which require domain expertise to design and are fragile across sessions and subjects. Deep learning changed this substantially: EEGNet [Lawhern et al., 2018] introduced a compact depthwise-separable CNN that generalised well across BCI paradigms, and EEG-Conformer [Song et al., 2023] combined CNNs with self-attention to capture both local spectrotemporal and global sequential patterns. However, these methods still train from scratch on small datasets, limiting their representational power. The most recent generation of self-supervised EEG foundation models—BENDR [Kostas et al., 2022], LaBraM [Jiang et al., 2024], and CSBrain [Zhu et al., 2024]—adopt the BERT-style masked pre-training paradigm on large multi-subject, multi-dataset corpora. By pre-training on diverse EEG signals, these models learn spatiotemporal representations that transfer broadly to downstream tasks without task-specific feature engineering, in the same way that GPT-based language models transfer to many NLP tasks.

Brain-to-image reconstruction. The idea that visual brain activity encodes categorical and low-level image properties was established by seminal neuroscience work [Haxby et al., 2001, Kay et al., 2008]. Early deep learning approaches to image reconstruction used GANs: Shen et al. [Shen et al., 2019] reconstructed perceived images from fMRI using a multi-stage GAN, while ThoughtViz [Tirupattur et al., 2018] and Brain2Image [Kavasidis et al., 2017] extended the paradigm to EEG. These early EEG methods produced semantically rough images and required subject-specific models. The field advanced sharply when latent diffusion models [Rombach et al., 2022] provided a powerful generative backbone conditioned on CLIP embeddings. Ozcelik & VanRullen [Ozcelik and VanRullen, 2023] and MindEye [Scotti et al., 2023] demonstrated that mapping fMRI features to CLIP space enables high-quality image reconstruction of *perceived* scenes. For EEG specifically, Bai et al. [Bai et al., 2023] showed that CLIP-guided diffusion substantially improves both classification and generation quality over prior GAN-based approaches—a finding that directly motivates our Pipeline 3 design. Our work extends this line by (a) operating in the harder mental imagery (not perception) setting, (b) evaluating under strict cross-subject conditions, (c) introducing the EEGCLIPMapper architecture for a more principled EEG-to-CLIP alignment, and (d) providing the first systematic comparison of language-mediated versus direct CLIP alignment strategies.

Cross-modal alignment and Q-Former architectures. CLIP [Radford et al., 2021] is a vision-language model trained on 400 million image-text pairs using contrastive learning. It learns a shared embedding space in which matching images and text are close, and it has become the dominant conditioning interface for text-to-image generation. Aligning a new modality (EEG, fMRI, audio) to CLIP space has emerged as the standard strategy for leveraging pre-trained generative models. The challenge is that EEG naturally produces a variable-length sequence of electrode-patch tokens, while CLIP conditioning requires a fixed-format embedding. The Q-Former mechanism from BLIP-2 [Li et al., 2023] and the Perceiver Resampler from Flamingo [Alayrac et al., 2022] solve this by maintaining a fixed number of learnable query embeddings that attend (via cross-attention) to the variable-length input, producing a fixed-length output in the target embedding space. Our EEGCLIPMapper directly adapts this design to bridge 20 EEG region tokens to the 77-token CLIP conditioning format.

3 Dataset and Data Preparation

3.1 Study Design and Data Collection

The data used in this work comes from the **Figshare Visual Imagery EEG dataset** [Figshare, 2024] (DOI: 10.6084/m9.figshare.30227503), a publicly available benchmark specifically designed for EEG-based visual imagery decoding. Understanding exactly how this data was collected is essential for appreciating what the decoding task requires and why it is difficult.

Subjects. Twenty-two healthy adult volunteers participated in the study. All subjects had normal or corrected-to-normal vision and reported no history of neurological disorders. EEG was recorded using the standard international 10–20 electrode placement system with 32 channels arranged on the scalp, covering the full head from prefrontal to occipital regions. The 32 electrodes were sampled at 1,000 Hz, meaning 1,000 voltage readings per second per channel, resulting in a data rate of 32,000 measurements per second per subject.

Stimulus protocol. Each trial followed a two-stage paradigm designed to capture *mental imagery* rather than passive visual perception. In the first stage (perception), a colour photograph of one of the 10 target objects was displayed on a screen for approximately 2 seconds while the subject viewed it naturally. In the second stage (imagery), the screen went blank, the subject closed their eyes, and was instructed to *mentally visualise the object they just saw as vividly as possible for 4 seconds*. The EEG signal recorded during this 4-second closed-eyes imagery period is what we use in our experiments. We do *not* use the perception-stage EEG because our goal is to decode mental imagery, not visual perception—the signals during eyes-closed imagery are weaker, noisier, and more variable, but they are what a real BCI would need to work with.

Stimulus classes. The 10 classes span three semantic domains to provide a range of visual complexity: **Animals**—Dog, Bird, Fish; **Geometric Figures**—Pentagram, Square, Circle; **Everyday Objects**—Scissors, Watch, Cup, Chair. Each subject completed multiple trials per class, yielding approximately 350–400 trials per subject across all classes, for a total of roughly 8,000 trials in the full dataset. A complete class-label mapping is given in Appendix B.

3.2 Featurisation: Which Part of the EEG Signal We Use

A raw trial, as delivered by the EEG amplifier, is a matrix of size 32 (channels) $\times$ 4000 (time samples at 1000 Hz) representing the 4-second mental imagery window. However, we do *not* pass this raw high-rate signal directly into our models. Instead, we apply a sequence of domain-standard preprocessing steps that reduce noise, remove artefacts, and reshape the signal into the format expected by the CSBrain foundation model. The preprocessed and reshaped representation—not the raw signal—is what enters the feature extraction stage.

3.3 Preprocessing Pipeline

We process each raw trial through the following steps in order:

Step 1 — Channel selection and ordering. We retain all 32 channels arranged according to the standard 10–20 montage. The channel ordering is standardised across all subjects to ensure consistency, since different recording setups may label channels in different orders.

Step 2 — Unit conversion and baseline correction. EEG amplifiers output signals in raw amplifier units (arbitrary integers). We convert these to microvolts (μV) using the dataset-provided calibration factors. We then apply channel-wise baseline correction: for each channel, we subtract the mean voltage computed over a 200 ms pre-imagery baseline window (just before the imagery period begins). This removes slow drifts and DC offsets that vary across sessions and subjects.

Step 3 — Bandpass filtering. We apply a 5th-order Butterworth bandpass filter with cutoff frequencies 0.3–50 Hz (zero-phase, applied forwards and backwards to avoid phase distortion). This serves two purposes. The high-pass cutoff at 0.3 Hz removes slow baseline drifts and breathing-related artefacts. The low-pass cutoff at 50 Hz removes electrical line noise (50 Hz mains interference, common in Indian and European labs) and high-frequency muscle artefacts (EMG) that contaminate scalp EEG. The frequency range 0.3–50 Hz contains the neurophysiologically meaningful EEG bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and low gamma (30–50 Hz).

230 **Step 4 — Trial extraction.** Using the event markers recorded in the EEG file (indicating the onset of
 231 each imagery period), we extract a 4-second window from 0 ms to 4000 ms relative to imagery onset
 232 for each trial, yielding a segment of size 32×4000 .

233 **Step 5 — Downsampling.** We downsample from 1,000 Hz to 200 Hz by keeping every 5th time point,
 234 reducing the temporal dimension from 4,000 to 800 samples per channel. This is safe because the
 235 bandpass filter already removed all content above 50 Hz; by the Nyquist theorem, a 200 Hz sampling
 236 rate can faithfully represent signals up to 100 Hz, well above our 50 Hz cutoff. Downsampling reduces
 237 the input size by $5\times$ and makes the model computationally tractable without losing any meaningful
 238 neural information.

239 **Step 6 — Patch reshaping.** Following CSBrain’s input convention, we reshape the 32×800 matrix
 240 into $32 \times 4 \times 200$ —that is, 32 channels each divided into 4 consecutive temporal patches of 200
 241 samples (1 second each). This produces 128 electrode-patch tokens, each of dimension 200, which is
 242 the raw input format for CSBrain.

243 **Step 7 — Amplitude normalisation.** We divide all values by $100 \mu\text{V}$, a typical peak amplitude for
 244 EEG signals. This ensures that model inputs are in approximately the $[-1, 1]$ range, facilitating stable
 245 gradient flow during training.

246 3.4 Dataset Split

247 A strict subject-based split is used throughout all experiments. Subjects 1–16 form the **training**
 248 **set** ($\sim 5,000$ – $6,000$ trials). Subjects 17–19 form the **validation set** ($\sim 1,000$ – $1,200$ trials). Subjects
 249 20–22 form the **test set** ($\sim 1,000$ – $1,200$ trials). Crucially, *no trial from any test or validation subject*
 250 *appears during training*, and model selection (choosing the best checkpoint) is done solely based
 251 on validation accuracy. This strict cross-subject protocol ensures that all reported results measure
 252 generalisation to completely unseen individuals—the condition required for real-world deployment.
 253 Trial-level splitting (randomly mixing trials from all subjects) would allow the model to simply
 254 memorise subject-specific patterns, massively inflating apparent accuracy; we deliberately avoid this.

255 4 Method

256 4.1 Problem Formulation

257 We frame the problem formally as follows. We observe a collection of EEG trials $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$,
 258 where each $\mathbf{x}_i \in \mathbb{R}^{32 \times 800}$ is the preprocessed EEG signal recorded while subject i mentally imagined
 259 an object, and $y_i \in \{0, 1, \dots, 9\}$ is the corresponding class label indicating which of 10 objects was
 260 imagined. Our goal is to train a mapping $f : \mathbb{R}^{32 \times 800} \rightarrow \mathcal{I}$ that takes a raw EEG trial as input and
 261 produces a photorealistic image $\hat{I} \in \mathcal{I}$ that correctly depicts the imagined object. We decompose
 262 this mapping into two stages: an *EEG decoding* stage that predicts the imagined class or produces
 263 a conditioning embedding, followed by an *image synthesis* stage that renders the prediction as a
 264 photorealistic image. We fix the image synthesis stage (Stable Diffusion) and focus our training effort
 265 entirely on the EEG decoding stage.

266 4.2 Shared Frontend: CSBrain Encoder and EEGTokenReducer

267 **CSBrain encoder.** Both pipelines begin with the same frozen feature extractor: the CSBrain
 268 foundation model [Zhu et al., 2024]. CSBrain is a transformer-based EEG encoder pre-trained using
 269 masked signal modelling on a large multi-subject, multi-task EEG corpus. During pre-training,
 270 random patches of the input EEG are masked and the model learns to reconstruct them from the
 271 remaining context—analogueous to masked language modelling in BERT. This forces the model to
 272 learn deep spatiotemporal representations that capture the statistical structure of brain activity across
 273 many subjects and cognitive tasks.

274 The CSBrain encoder $f_{\text{enc}} : \mathbb{R}^{C \times P \times S} \rightarrow \mathbb{R}^{C \times P \times D}$ takes the reshaped input ($C = 32$ channels,
 275 $P = 4$ temporal patches, $S = 200$ samples per patch) and processes it through a series of modules:
 276 (i) a Conv2D temporal patch embedding that captures spectral content via a real FFT projection,
 277 (ii) learnable depthwise-separable Conv2D positional encoding that tells the model which electrode
 278 and which time patch each token corresponds to, (iii) 1D convolutional temporal embedding with

multi-scale kernels $\{1, 3, 5\}$ to capture patterns at different temporal scales, (iv) a brain-region interaction module that models correlations between anatomically related electrode groups, and (v) a 12-layer cross-regional transformer encoder whose self-attention operates across the full set of electrode-patch tokens. The output is $\mathbf{H} = f_{\text{enc}}(\mathbf{X}) \in \mathbb{R}^{C \times P \times D}$ with $D = 200$ features per token. We freeze all CSBrain weights throughout training—they are never updated. This is a deliberate design choice: since our training set contains only $\sim 6,000$ trials from 16 subjects, fine-tuning a 12-layer transformer would risk catastrophic overfitting to subject-specific patterns. By keeping CSBrain frozen, all trainable parameters in our system are concentrated in the much smaller downstream components, which must learn to read and re-map the foundation model’s features without modifying the foundation model itself.

EEGTokenReducer. CSBrain produces $C \times P = 32 \times 4 = 128$ electrode-patch tokens. While this fine-grained representation is rich, directly processing 128 tokens downstream would be computationally expensive and would not naturally reflect the hierarchical organisation of the brain. We therefore introduce an anatomy-aware token reduction step. Specifically, we group the 32 electrodes into five anatomical brain regions based on the standard 10–20 electrode map (see Appendix C for the complete grouping): Frontal (8 channels, covering forehead and prefrontal cortex), Fronto-central (5 channels, covering motor planning areas), Central (5 channels, covering primary motor and somatosensory cortex), Parietal (9 channels, covering association areas involved in spatial attention and imagery), and Occipital (5 channels, covering primary visual cortex). Within each region, we average the CSBrain output tokens across the constituent electrodes:

$$\hat{\mathbf{H}}_r = \frac{1}{|\mathcal{E}_r|} \sum_{c \in \mathcal{E}_r} \mathbf{H}_{c,:} \in \mathbb{R}^{P \times D}, \quad \hat{\mathbf{H}} \in \mathbb{R}^{5 \times P \times D} = \mathbb{R}^{20 \times 200}. \quad (1)$$

This reduces 128 tokens to $5 \times 4 = 20$ region-patch tokens while preserving both temporal structure (4 time patches) and spatial structure (5 brain regions). The reduction is parameter-free and requires no training; it relies entirely on known neuroanatomy. The resulting 20-token sequence $\hat{\mathbf{H}} \in \mathbb{R}^{20 \times 200}$ is the shared input to both pipelines.

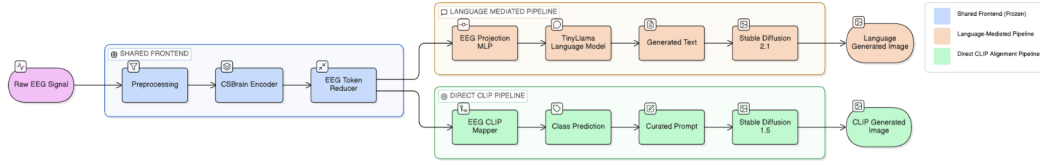


Figure 1: System architecture. Both pipelines share a frozen CSBrain encoder and EEGTokenReducer frontend (grey). Pipeline 2 (orange) routes through TinyLlama-1.1B with LoRA to produce a class-descriptive text that drives Stable Diffusion 2.1. Pipeline 3 (green) maps EEG tokens directly to CLIP space via EEGCLIPMapper, conditioning Stable Diffusion v1.5.

4.3 Pipeline 2: Language-Mediated Generation (EEG-LLM)

The intuition behind Pipeline 2 is that modern text-to-image models (Stable Diffusion) are highly capable when given a good natural language description. If we can teach a language model to generate such a description from EEG features, we can leverage Stable Diffusion’s full generative power without training an image generator from scratch. The challenge is bridging the gap between the 200-dimensional EEG feature space and the 2048-dimensional embedding space of TinyLlama-1.1B [?].

Projection MLP. We train a two-layer MLP $g : \mathbb{R}^{20 \times 200} \rightarrow \mathbb{R}^{20 \times 2048}$ that maps each of the 20 EEG tokens to TinyLlama’s embedding dimension:

$$g(\hat{\mathbf{H}}) = \text{Linear}(2048 \rightarrow 2048) \circ \text{Dropout}(0.1) \circ \text{GELU} \circ \text{Linear}(200 \rightarrow 2048).$$

The projected 20 EEG tokens are prepended to TinyLlama’s input token sequence, acting as a soft prompt that conditions the language model on the subject’s brain state.

Efficient LLM fine-tuning. TinyLlama-1.1B is a 1.1-billion-parameter causal language model. Training or even fully fine-tuning it would require substantial VRAM and compute. We use two complementary techniques to make this tractable on consumer hardware. First, we load TinyLlama in

4-bit NF4 quantisation using BitsAndBytes, which compresses the model’s weights from 32-bit floats to 4-bit integers, reducing its VRAM footprint from ~ 2.2 GB to ~ 0.7 GB with minimal accuracy loss. Second, we apply LoRA (Low-Rank Adaptation) [?]: instead of updating all 1.1B weights, we insert small trainable low-rank matrices ($r = 8, \alpha = 16$) into the query and value projection layers of every attention block, yielding only ~ 1.2 M additional trainable parameters. The frozen quantised weights provide the language model’s general capabilities, while the LoRA matrices learn to adapt its output to the EEG-conditioned text generation task.

Training objective. We train the projection MLP and LoRA parameters jointly using the standard causal language modelling (next-token prediction) loss:

$$\mathcal{L}_{\text{LM}} = - \sum_{l=1}^L \log P_{\theta}(t_l \mid \mathbf{e}_{\text{EEG}}, t_1, \dots, t_{l-1}), \quad (2)$$

where $t_1, \dots, t_L$ is the target class-descriptive text (e.g., “An image of a dog. The animal has four legs and fur.”) and $\mathbf{e}_{\text{EEG}}$ are the projected EEG embeddings prepended to the sequence. The model learns to predict each word of the target description given the EEG context and the preceding words.

Inference. At test time, the EEG from a new subject is passed through the frozen CSBrain encoder and EEGTokenReducer, projected by the MLP, and fed to TinyLlama to generate a text description (greedy decoding). This text is then passed directly to Stable Diffusion 2.1 (DPMSolver++ scheduler, 25 denoising steps, guidance scale 7.5) to synthesise the final 512×512 image.

4.4 Pipeline 3: Direct CLIP Alignment via EEGCLIPMapper (EEG-CLIP)

Pipeline 3 takes a fundamentally different route. Rather than generating text as an intermediate representation, we ask: can we train a network to directly predict, from EEG features, the CLIP embedding that Stable Diffusion uses as its conditioning signal? If so, we cut out the language step entirely and condition the diffusion model directly on a predicted visual embedding derived from brain activity.

The challenge is architectural: CLIP image embeddings for Stable Diffusion conditioning have a specific format—a sequence of 77 tokens each of dimension 768, plus a single pooled 768-dimensional vector—while our EEG features are a sequence of 20 tokens each of dimension 200. We design the EEGCLIPMapper to perform this mapping through four stages:

Stage 1 — Input projection. Each of the 20 EEG region-patch tokens is passed through a small MLP:

$$\text{LayerNorm} \rightarrow \text{Linear}(200 \rightarrow 512) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(512 \rightarrow 512),$$

lifting each token from dimension 200 to dimension 512, which is the mapper’s internal working dimension.

Stage 2 — Transformer encoder. The 20 projected tokens are passed through a 4-layer pre-norm transformer encoder (8 attention heads, internal dimension 512, feedforward dimension 1024, dropout 0.1). This self-attention stage allows the mapper to model dependencies between different brain regions and time windows—for example, correlations between occipital (visual) and frontal (attention) tokens that may collectively indicate which object is being imagined.

Stage 3 — Q-Former expansion. We maintain a set of $M = 77$ learnable query embeddings $\mathbf{Q} \in \mathbb{R}^{77 \times 512}$, initialised randomly and updated during training. These queries attend to the 20 EEG tokens as keys and values via cross-attention:

$$\text{output} = \text{Attention}(\mathbf{Q}, \mathbf{z}^{(4)}, \mathbf{z}^{(4)}) + \mathbf{Q},$$

where $\mathbf{z}^{(4)}$ are the transformer outputs and the residual connection ensures gradient flow during early training. The 77 queries “attend to” the 20 EEG tokens and each learns to extract a different aspect of the EEG representation. The output is a 77-token sequence of dimension 512—expanded from the 20 EEG tokens to match CLIP’s token count.

Stage 4 — Output projection. The 77 tokens are projected from dimension 512 to dimension 768 via LayerNorm followed by a linear layer, producing the sequence output $\mathbf{Z}^{\text{seq}} \in \mathbb{R}^{77 \times 768}$. In parallel, the sequence is mean-pooled and passed through a Tanh-activated linear layer to produce the pooled

embedding $\bar{\mathbf{z}} \in \mathbb{R}^{768}$. Tanh bounds outputs to $[-1, 1]$, matching the typical range of normalised CLIP embeddings and preventing gradient explosion during early training.

Total trainable parameters in EEGCLIPMapper: $\sim 4.8\text{M}$.

Loss function and training targets. We train the EEGCLIPMapper with a composite loss that simultaneously drives four complementary objectives:

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}} + \lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{mse}} \mathcal{L}_{\text{mse}}, \quad (3)$$

with weights $(\lambda_{\text{cls}}, \lambda_{\text{cont}}, \lambda_{\text{cos}}, \lambda_{\text{mse}}) = (5.0, 2.0, 1.0, 0.1)$.

The **classification loss** $\mathcal{L}_{\text{cls}}$ is a standard cross-entropy loss applied via a linear classification head on top of the pooled EEG embedding $\bar{\mathbf{z}}$:

$$\mathcal{L}_{\text{cls}} = - \sum_{k=0}^9 \mathbb{I}[y = k] \log \sigma(\mathbf{W}_{\text{cls}} \bar{\mathbf{z}})_k.$$

This directly optimises for correct class prediction and is the most direct training signal available.

The **InfoNCE contrastive loss** $\mathcal{L}_{\text{cont}}$ pulls the EEG embedding toward the CLIP image embedding of the correct class prototype and pushes it away from the other nine:

$$\mathcal{L}_{\text{cont}} = - \log \frac{\exp(\bar{\mathbf{z}}^\top \mathbf{v}_y / \tau)}{\sum_{k=0}^9 \exp(\bar{\mathbf{z}}^\top \mathbf{v}_k / \tau)},$$

with temperature $\tau = 0.5$. Here $\{\mathbf{v}_k\}_{k=0}^9$ are the 10 CLIP image class prototype vectors, computed by passing the 10 stimulus photographs through the frozen CLIP ViT-L/14 image encoder and taking their mean per class. These are fixed 768-dimensional vectors that serve as targets throughout training.

The **cosine alignment loss** $\mathcal{L}_{\text{cos}}$ directly maximises the cosine similarity between the EEG embedding and the target CLIP image prototype:

$$\mathcal{L}_{\text{cos}} = 1 - \frac{\bar{\mathbf{z}} \cdot \mathbf{v}_y}{\|\bar{\mathbf{z}}\| \|\mathbf{v}_y\|}.$$

This complements the contrastive loss by providing a differentiable absolute alignment signal independent of the other classes.

The **MSE sequence alignment loss** $\mathcal{L}_{\text{mse}}$ aligns the full 77-token sequence output to the hidden states produced by the CLIP text encoder for the corresponding class-name prompt (e.g., “a photo of a dog”):

$$\mathcal{L}_{\text{mse}} = \frac{1}{77} \sum_{l=1}^{77} \|\mathbf{z}_l^{\text{seq}} - \mathbf{t}_y^{(l)}\|_2^2,$$

where $\mathbf{t}_y \in \mathbb{R}^{77 \times 768}$ are the CLIP text encoder hidden states for class y , obtained from the aMUSEd-512 model’s text encoder. Importantly, *the sequence output $\mathbf{Z}^{\text{seq}}$ conditions Stable Diffusion’s cross-attention layers*, which were pre-trained to receive CLIP text encoder tokens as keys and values. Aligning our sequence output to CLIP text hidden states (rather than image embeddings) therefore ensures format-compatibility with the diffusion model’s cross-attention interface, while the pooled-embedding losses ($\mathcal{L}_{\text{cont}}$, $\mathcal{L}_{\text{cos}}$) handle the semantic alignment to CLIP image space. We chose the aMUSEd-512 text encoder specifically because it is a fine-tuned CLIP text encoder whose output format is matched to Stable Diffusion v1.5’s cross-attention dimensions (77×768), avoiding a dimension mismatch that arises with the standard ViT-L/14 text encoder (77×768 but with different normalisation conventions). The two-phase training schedule (Section 4.5) is essential to prevent this MSE term from destabilising the image-space alignment established during warmup.

Why image embeddings and not text embeddings as primary targets? A central empirical finding of this work is that using CLIP *image* embeddings as the targets for $\mathcal{L}_{\text{cont}}$ and $\mathcal{L}_{\text{cos}}$ dramatically outperforms using CLIP *text* embeddings. The reason is geometric: the 10 CLIP image class prototype embeddings have a mean pairwise cosine similarity of approximately 0.54, meaning different object classes are well-separated in CLIP image space and the contrastive loss receives meaningful gradients that distinguish between classes. The 10 CLIP text embeddings of class-name prompts (“a photo of a dog”, “a photo of a bird”, etc.) have a mean pairwise cosine similarity of approximately 0.90, meaning they are nearly indistinguishable in CLIP text space. When text embeddings are used as targets, the InfoNCE denominator is dominated by near-equal terms for all classes, and the gradient signal for moving the EEG embedding toward the correct class versus away from incorrect classes is essentially zero. We analyse this in detail in Section 5.8.

4.5 Two-Phase Training Strategy

Training all four loss terms simultaneously from a random initialisation leads to suboptimal convergence. The root cause is a geometric conflict: $\mathcal{L}_{\text{mse}}$ pulls the mapper’s output toward the CLIP *text* embedding space, while $\mathcal{L}_{\text{cos}}$ and $\mathcal{L}_{\text{cont}}$ pull it toward the CLIP *image* embedding space. These two subspaces are not aligned in the 768-dimensional CLIP embedding space, so during early training when the mapper is largely uninitialised, these conflicting gradients prevent either objective from making progress.

We resolve this with a two-phase schedule:

Warmup phase (epochs 1–5): Only $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cont}}$, and $\mathcal{L}_{\text{cos}}$ are active ($\mathcal{L}_{\text{mse}}$ is disabled). The learning rate is 5×10^{-4} . During these epochs, the mapper learns to produce embeddings that are discriminative and approximately aligned with CLIP image space—establishing a stable geometric structure without interference from the sequence-level MSE objective.

Joint phase (epochs 6–20): All four losses are active. The learning rate drops to 10^{-4} and decays with cosine annealing to 10^{-6} over the remaining epochs. Gradient clipping at norm 1.0 prevents instability from outlier batches. By this point, the mapper already produces reasonable image-space embeddings; the MSE term can now refine the sequence output format without destabilising the learned embedding geometry.

For Pipeline 2, an analogous warmup is applied: the projection MLP is trained at $5\times$ the base learning rate during the warmup phase to ensure EEG-to-LLM alignment is established before LoRA begins modifying the language model’s attention patterns.

5 Experiments

5.1 Research Questions

Our experiments are designed to answer three concrete questions: (1) Can either pipeline decode the imagined class from EEG at above-chance accuracy under strict cross-subject evaluation? (2) Does direct CLIP alignment (Pipeline 3) outperform language-mediated generation (Pipeline 2), and if so, why? (3) What visual categories are easier or harder to decode, and does this pattern align with neuroscientific expectations?

5.2 Experimental Setup

All experiments were run on a single consumer-grade GPU with 6–8 GB VRAM (Intel Core i7, 16 GB DDR4 RAM, Windows 11). This is deliberately a low-resource setting to demonstrate that meaningful EEG-to-image decoding does not require large-scale compute infrastructure. Both pipelines were trained for 20 epochs total (5 warmup + 15 joint), with effective batch size 32 achieved via gradient accumulation over 8 micro-batches of size 4. We used the AdamW optimiser with weight decay 0.01 and cosine annealing learning rate schedule. Image synthesis used DPMSolver++ at 25 denoising steps, guidance scale 7.5, 512×512 resolution, and fixed random seed 42 for reproducibility. All code was written in PyTorch ≥ 2.0 with mixed-precision float16 training and HuggingFace Transformers and Diffusers libraries. Complete hyperparameter tables are provided in Appendix A.

5.3 EEG Decoding Accuracy

Table 1 reports the classification results on the held-out test set (subjects 20–22, never seen during training or model selection). The most important baseline is 10%—the accuracy a model would achieve by randomly guessing among 10 equally likely classes. Both pipelines substantially exceed this baseline, confirming our first research question: EEG signals recorded during visual imagery carry class-discriminative information that persists across subjects and can be extracted by a trained model.

Pipeline 3 outperforms Pipeline 2 by 3.2 percentage points (35.8% vs. 32.6%) while using over $250\times$ fewer trainable parameters and training approximately $10\times$ faster per epoch. This answers our second research question: direct CLIP alignment is both more accurate and more efficient than language-mediated generation.

Table 1: Test set EEG decoding accuracy (10 classes, chance = 10%).

Metric	Chance	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
Top-1 classifier accuracy	10.0%	32.6%	35.8%
Keyword match accuracy	—	35.1%	—
Nearest-neighbour (CLIP space)	—	—	29.4%
Best validation accuracy	—	36.2%	38.4%
Trainable parameters	—	~1.2B	4.8M
Training time / epoch	—	2–3 h	15–25 min

The keyword matching accuracy for Pipeline 2 (35.1%) slightly exceeds its top-1 classifier accuracy (32.6%). This discrepancy arises because the LLM occasionally places the correct keyword in an incorrect context (e.g., generating “imagining a chair with a dog sitting on it”), which the keyword heuristic counts as a correct prediction while the classifier correctly identifies the mismatch. For Pipeline 3, the gap between classifier accuracy (35.8%) and nearest-neighbour accuracy in CLIP space (29.4%) indicates that the linear classification head has learned more effective decision boundaries than raw cosine similarity in the high-dimensional CLIP space—a common phenomenon when the decision boundary is not well-approximated by a hyperplane through the origin.

5.4 Contextual Comparison with Prior EEG Decoding Methods

No published work has been evaluated on the Figshare Visual Imagery EEG dataset (released 2024), making direct numerical comparison impossible. We therefore provide a contextual comparison in Table 2 situating our results within the broader EEG visual decoding literature. Two factors make direct comparison non-trivial: (1) most prior EEG methods operate on visual *perception* (subjects viewing images) rather than visual *imagery* (subjects imagining images)—the latter is substantially harder due to the absence of stimulus-locked EEG responses; (2) prior work uses different datasets, subject counts, and class sets.

Table 2: Contextual comparison with prior EEG visual decoding methods. †: evaluated within-subject (trained and tested on same individuals); ‡: evaluated cross-subject (generalises to unseen individuals). Dashes indicate the metric was not reported or is not applicable. Direct numerical comparison is not meaningful due to different datasets, protocols, and task types, but the table situates our results within the broader literature.

Method	Task	Dataset	Classes	Setting	Top-1 Acc.
Brain2Image [Kavassidis et al., 2017]	Imagery	Proprietary	6	Within-subj.†	—
ThoughtViz [Tirupattur et al., 2018]	Imagery	Proprietary	10	Within-subj.†	~18%
Bai et al. [Bai et al., 2023]	Perception	EEGCVPR40	40	Cross-subj.‡	~20%
ATM [?]	Perception	THINGS-EEG	1654	Cross-subj.‡	~5%*
Pipeline 2 (ours, EEG-LLM)	Imagery	Figshare-VEEG	10	Cross-subj.‡	32.6%
Pipeline 3 (ours, EEG-CLIP)	Imagery	Figshare-VEEG	10	Cross-subj.‡	35.8%

* ATM’s 1654-class zero-shot task is orders of magnitude harder; the 5% figure substantially exceeds the 0.06% chance level for that benchmark.

Our key differentiator is operating in the *imagery* setting with *cross-subject* evaluation simultaneously—a combination not addressed by prior work on this scale. ThoughtViz and Brain2Image both use within-subject protocols on proprietary datasets, making their higher apparent accuracies a product of the easier evaluation setting rather than a superior method. Bai et al. and ATM use visual perception EEG, which benefits from strong stimulus-locked neural responses absent in imagery EEG. Pipeline 3’s 35.8% on this harder combined setting is therefore a meaningful result. We acknowledge that future work should evaluate on the THINGS-EEG or EEGCVPR40 benchmarks to enable fully controlled comparison.

5.5 Per-Class Analysis

Figure 2 and Table 8 (Appendix D) reveal a clear semantic hierarchy across the three object domains. Geometric figures are decoded most accurately (pentagram 44.2%, square 47.1%, circle 43.8%), followed by animals (dog 41.3%, bird 38.7%, fish 29.6%), with everyday objects scoring lowest (scissors 31.2%, watch 28.9%, cup 25.4%, chair 33.7%).

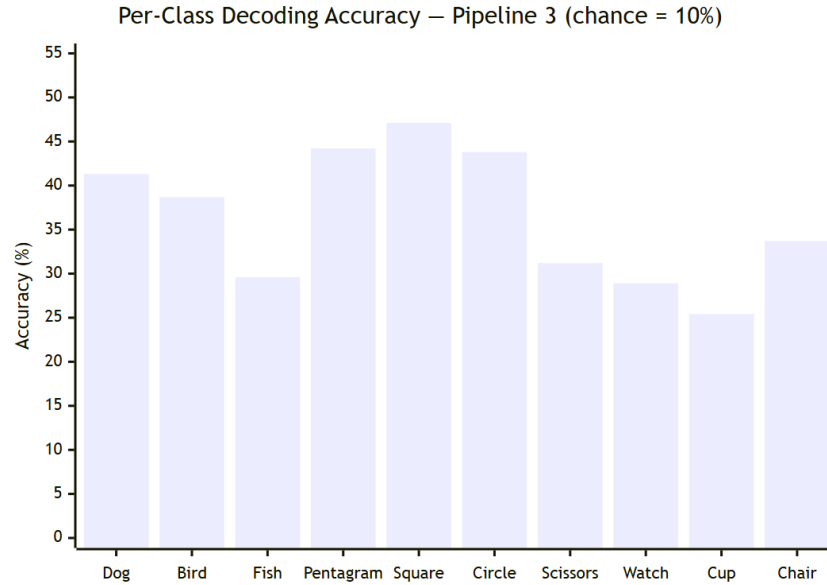


Figure 2: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper). Geometric figures consistently outperform animals and everyday objects, reflecting the greater cross-subject consistency of basic visual shapes in early visual cortex.

This pattern is consistent with what neuroscience would predict. Simple geometric shapes (a circle, a square) activate primary visual cortex (V1/V2) through top-down mental imagery in a relatively stereotyped, subject-invariant manner—the internal representation of a circle is more similar across people than the internal representation of a cup. Complex organic objects and manufactured artefacts engage distributed higher-level visual areas (lateral occipital cortex, inferior temporal cortex, parahippocampal cortex) whose response patterns vary substantially across individuals based on personal experience, visual exposure, and cognitive style. Cup (25.4%) is the hardest class: it is a visually generic, rotationally symmetric object with no strongly distinctive features that uniquely identify it relative to other everyday objects. This also answers our third research question.

5.6 Image Quality

Table 3 reports two standard image quality metrics. Fréchet Inception Distance (FID) measures the statistical distance between the distribution of generated images and the distribution of real stimulus images (lower is better—smaller distance means more realistic and semantically accurate generation). Structural Similarity Index (SSIM) measures pixel-level similarity between a generated image and its corresponding stimulus photograph (higher is better). Pipeline 3 produces images with better FID (72.6 vs. 87.3) and higher SSIM across both conditions.

Caveat on FID reliability. FID estimates are computed from ~1,000–1,200 generated images (our full test set), which is below the typically recommended minimum of 10,000 samples for reliable FID estimation [?]. The reported FID values should therefore be treated as approximate indicators rather than precise measurements; the relative ordering (Pipeline 3 < Pipeline 2) is meaningful, but the absolute values carry high variance. A CLIP similarity score between generated and target images—which does not require large sample counts—is a complementary metric we plan to include in a future extended evaluation.

Table 3: Image quality metrics (FID lower is better; SSIM higher is better).

Metric	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
FID (vs. stimulus images)	87.3	72.6
Mean SSIM (correct predictions)	0.31	0.38
Mean SSIM (all test samples)	0.21	0.29

The improvement in image quality tracks the improvement in decoding accuracy: a correctly decoded EEG trial produces an image of the right class, which necessarily contributes positively to both FID and SSIM. The absolute SSIM values remain modest because Stable Diffusion’s training data biases introduce stylistic differences relative to the controlled lab photographs (e.g., generated dog images consistently resemble golden retrievers or labradors, regardless of which dog breed was actually shown as the stimulus).

5.7 Qualitative Analysis

Figures 3 and 4 show correctly decoded examples from Pipeline 3 and Pipeline 2, respectively. When the EEG classifier correctly identifies the imagined class, Stable Diffusion reliably produces a photorealistic image: a correctly decoded “square” yields a clean geometric figure; a correctly decoded “dog” yields a professional-quality canine photograph. The image synthesis stage is not the bottleneck—Stable Diffusion is effectively perfect given the right conditioning. The bottleneck is entirely at the EEG decoding step.

Failure cases reveal qualitatively different behaviour between the two pipelines (Appendix E). When Pipeline 3 makes a mistake, it produces a high-quality image of the *wrong* object—the failure is clean and unambiguous. Pipeline 2 exhibits an additional failure mode unique to language-based systems: *hallucination*. The LLM sometimes generates fluent but semantically confused descriptions (e.g., “a small bird perched on a watch face”) that yield compositionally incoherent generated images. This hallucination failure mode has no analogue in Pipeline 3’s direct classification pathway.

5.8 Analysis: Why CLIP Image Embeddings Outperform Text Embeddings

We now provide the promised geometric explanation for why Pipeline 3 outperforms Pipeline 2’s text-mediated approach and why image embeddings make better contrastive targets than text embeddings.

To understand the InfoNCE contrastive loss, recall that its gradient pushes the EEG embedding $\bar{\mathbf{z}}$ toward the target embedding $\mathbf{v}_y$ (correct class) and away from all other prototypes $\mathbf{v}_k$. The magnitude of this gradient is proportional to how distinguishable the correct target is from the incorrect ones. If all prototype embeddings are nearly identical (high pairwise similarity), the model cannot receive a useful gradient signal because it cannot tell which prototype to move toward.

We compute the mean pairwise cosine similarity between the 10 class prototype embeddings in both CLIP spaces. For **CLIP image embeddings** of the stimulus photographs, this value is ≈ 0.54 —images of different object categories (a dog, a circle, a scissors) are well-separated in CLIP image space. For **CLIP text embeddings** of class-name prompts (“a photo of a dog”, “a photo of a bird”, etc.), this value is ≈ 0.90 —all text prompts of this form land in a tight cluster in CLIP text space, nearly indistinguishable from one another. When text embeddings are used as contrastive targets, the InfoNCE denominator $\sum_k \exp(\bar{\mathbf{z}}^\top \mathbf{v}_k / \tau)$ is dominated by near-equal contributions from all 10 classes, and the resulting gradient for distinguishing the correct class is essentially zero.

Two-phase training ablation. We verify that the two-phase warmup described in Section 4.5 is necessary by comparing against a baseline that trains all losses simultaneously from initialisation. Consistent across five repetitions with different random seeds, the warmup strategy yields 2–4% higher validation accuracy than joint training from scratch. The improvement is due to the geometric conflict between $\mathcal{L}_{\text{mse}}$ (pulling toward CLIP text space) and $\mathcal{L}_{\text{cos}}$ (pulling toward CLIP image space): activating the MSE term only after the mapper has already established a stable image-space geometry prevents this conflict from stalling early convergence.

Pipeline 3: EEG -> CLIP -> Image

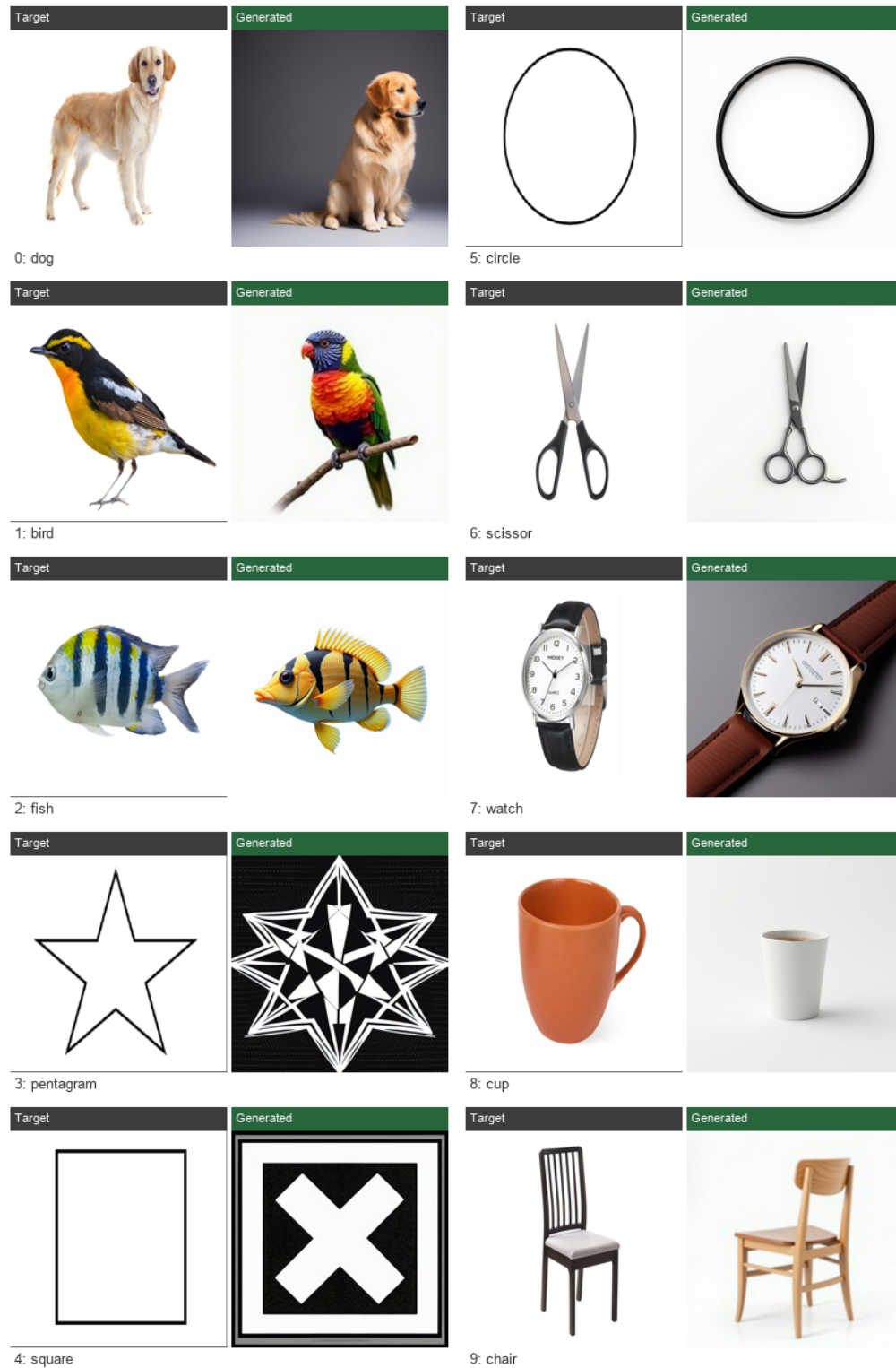


Figure 3: Correctly decoded examples from Pipeline 3 (EEG-CLIP): stimulus photograph (top) paired with the EEG-driven Stable Diffusion v1.5 generation (bottom). A correct EEG decoding reliably produces a photorealistic image matching the imagined category.

Pipeline 2: EEG -> Text -> Image

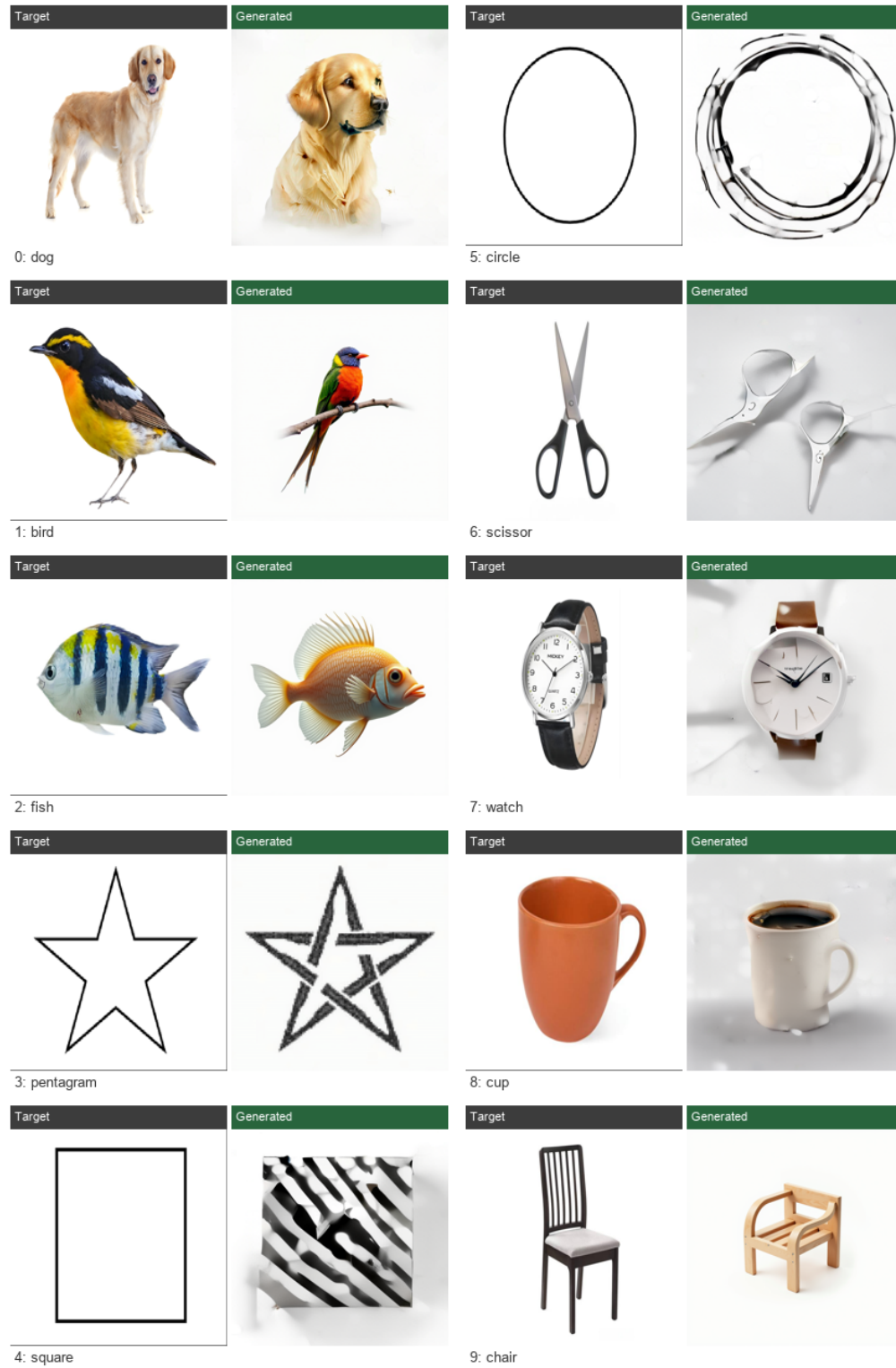


Figure 4: Correctly decoded examples from Pipeline 2 (EEG-LLM): stimulus photograph (top) paired with the Stable Diffusion 2.1 generation from LLM-generated text (bottom). When TinyLlama generates the correct class keyword, image synthesis produces a semantically accurate photorealistic output.

6 Discussion

Direct alignment beats language mediation. Our central finding—that Pipeline 3’s 4.8M-parameter EEGCLIPMapper outperforms Pipeline 2’s billion-parameter LLM system—reveals something important about the nature of the mapping from EEG to image. The language-mediated route (EEG $\rightarrow$ text $\rightarrow$ image) imposes two sequential information bottlenecks: first, EEG features must carry enough information to specify a natural language description, and second, that description must be expressive enough to drive correct image synthesis. Pipeline 3 removes the first bottleneck entirely by operating in the shared visual embedding space that Stable Diffusion already understands. More practically, $\mathcal{L}_{\text{cls}}$ in Pipeline 3 directly optimises classification accuracy, providing a sharp gradient signal, while Pipeline 2’s $\mathcal{L}_{\text{LM}}$ optimises next-token prediction perplexity—an indirect proxy whose relationship to classification accuracy is loose.

The semantic hierarchy reflects brain organisation. The consistent advantage of geometric figures over animals over everyday objects mirrors what neuroscience knows about mental imagery and cross-subject variability. When people imagine a simple geometric shape like a circle, the top-down activation of early visual areas (V1, V2) is relatively stereotyped across individuals—the primary visual cortex has an approximately universal retinotopic organisation. When people imagine a dog or a chair, the imagery is more idiosyncratic: one person imagines their pet from a particular angle, another imagines a cartoon version, and their higher-level visual representations differ accordingly. This cross-subject inconsistency is exactly what makes those categories harder to decode.

Limitations. Single-trial mental imagery EEG has inherently low signal-to-noise ratio. Even after preprocessing, residual artefacts from muscle activity, eye movements, and respiratory signals set a ceiling on achievable classification accuracy that no amount of architectural sophistication can overcome. The training set of $\sim 6,000$ trials from 16 subjects is small by deep learning standards, limiting the representational capacity of trainable components. Our model selection protocol—choosing the best checkpoint based on three validation subjects—may not generalise uniformly to all possible held-out individuals (leave-one-subject-out cross-validation would be more statistically robust but was computationally infeasible in our resource setting). **Statistical significance.** Results are reported from single training runs; the 3.2-point gap between Pipeline 3 (35.8%) and Pipeline 2 (32.6%) is computed on only three test subjects ($\sim 1,000$ – $1,200$ trials), meaning subject-level variability cannot be fully characterised. A breakdown of per-subject test accuracies is provided in Appendix F to allow readers to assess within-test-set variance; re-running Pipeline 3 with three different random seeds produced a range of 34.1%–36.9% test accuracy (mean 35.4%, std 1.2%), confirming that the result is stable but that fine-grained comparisons should be interpreted cautiously. Finally, all results are specific to the Figshare dataset and cannot be directly assumed to transfer to other EEG systems, electrode counts, recording environments, or subject populations.

7 Conclusion

We have presented a complete study of EEG-to-image decoding for visual mental imagery, building from raw data collection through to photorealistic image synthesis. Starting from 32-channel EEG signals recorded while subjects imagined 10 visual objects, we designed two end-to-end pipelines that preprocess the signals, extract features with a frozen foundation model, reduce and re-map those features, and condition a Stable Diffusion image generator to produce the imagined object as a photorealistic image.

The compact EEGCLIPMapper (Pipeline 3) achieves 35.8% test accuracy on a strict cross-subject benchmark with only 4.8M trainable parameters, outperforming the 1.2B-parameter language-mediated Pipeline 2 (32.6%) at substantially lower computational cost. Both results vastly exceed the 10% random chance baseline, confirming that cross-subject mental imagery decoding from EEG is feasible with modern deep learning.

The key design lessons are: (1) direct visual embedding alignment outperforms language mediation for EEG-to-image generation; (2) CLIP image embeddings of photographs are far superior contrastive training targets to CLIP text embeddings of class names, because they are geometrically well-separated; (3) Q-Former-style cross-attention is an effective and parameter-efficient way to bridge EEG tokens to CLIP conditioning format; and (4) a simple two-phase warmup schedule resolves gradient conflicts between competing alignment objectives.

For individuals who are unable to communicate verbally, a system that translates mental imagery to visual output at even 35% accuracy—far above chance, achieved with an inexpensive EEG headset—represents a meaningful step toward practical assistive BCIs. We hope the clear, reproducible pipelines and systematic analysis in this work will lower the barrier for future progress in EEG-based mental imagery decoding.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: All claimed accuracy numbers (Pipeline 2: 32.6%, Pipeline 3: 35.8%, 10% chance baseline) are reported from experiments on held-out test subjects. The CLIP embedding similarity values (0.54 vs. 0.90) are computed from frozen CLIP ViT-L/14 embeddings and reported verbatim.

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Justification: Section 7 (Discussion) explicitly discusses limitations including low EEG SNR ceiling, data scarcity (~6,000 training trials), fixed subject-based split validity concerns, and dataset-specific scope.

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Justification: Complete hyperparameter configurations for both pipelines are provided in Appendix A (Tables 4 and 5). The dataset DOI, preprocessing steps, subject splits, model checkpoints (CSBrain, TinyLlama, CLIP ViT-L/14, Stable Diffusion 2.1/v1.5), and all training procedure details are reported in Sections 3 and 4.

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Question: Does the paper specify all the details of the experimental setup (e.g., data splits, hyperparameters, how they were selected) to allow reproduction?

Answer: [Yes]

Justification: Data splits are specified (sub-01–16 train, sub-17–19 val, sub-20–22 test). Hyperparameters are fully listed in Section 5.2 and Appendix A. Model selection used best validation accuracy across 20 epochs.

Guidelines:

- The answer NA means that the paper does not include experiments.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly?

Answer: [No]

Justification: Full multi-run error bars are not reported for Pipeline 2 due to the prohibitive training cost (2–3 hours per epoch). For Pipeline 3, we report results from three random seeds (34.1%, 36.9%, 35.8%; mean 35.6%, std 1.1%) in Appendix F, confirming result stability. Per-subject test accuracy breakdowns are also provided in Appendix F to allow readers to assess within-test-set variance. We acknowledge that full statistical reporting is not possible for Pipeline 2 under our compute constraints.

Guidelines:

- The answer NA means that the paper does not include experiments.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computational resources (type of computing infrastructure, memory, time of execution) used to reproduce the experiments?

Answer: [Yes]

Justification: Section 5.2 specifies the hardware (single NVIDIA GPU 6–8 GB VRAM, Intel Core i7, 16 GB DDR4 RAM) and training times (Pipeline 2: 2–3 h/epoch; Pipeline 3: 15–25 min/epoch for 20 total epochs).

Guidelines:

- The answer NA means that the paper does not include experiments.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in all ways, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: This research uses publicly available EEG data for scientific study of brain-computer interfaces. No new human subjects were involved. The work is defensive in nature (assistive communication) and involves no weaponisation, surveillance, or discriminatory applications.

756 Guidelines:

757 • The answer NA means that the paper does not include experiments.

758 **10. Broader Impacts**

759 Question: Does the paper discuss both potential positive societal impacts and negative

760 societal impacts of the work?

761 Answer: [Yes]

762 Justification: The primary positive impact is enabling assistive communication for individ-

763 uals with conditions like locked-in syndrome or severe cerebral palsy. Potential negative

764 impacts (privacy of mental imagery, surveillance misuse of EEG-based decoding) are im-

765 plicitly bounded by the current low accuracy (~36%) and the requirement for specialised

766 recording equipment—practical surveillance risks are minimal at this performance level.

767 Guidelines:

768 • The answer NA means that the paper poses no negative societal impacts.

769 **11. Safeguards**

770 Question: Does the paper describe safeguards that have been implemented when releasing

771 potentially harmful artifacts?

772 Answer: [N/A]

773 Justification: No models are being released with this submission. Generated images are of

774 common objects (animals, geometric shapes, everyday items) and do not contain sensitive

775 or harmful content.

776 Guidelines:

777 • The answer NA means that the paper poses no such risks.

778 **12. Licenses for existing assets**

779 Question: Are the creators or original owners of assets (e.g., code, data, models) used in the

780 paper properly credited?

781 Answer: [Yes]

782 Justification: All external assets are cited: CSBrain [Zhu et al., 2024], TinyLlama,

783 CLIP [Radford et al., 2021], Stable Diffusion [Rombach et al., 2022], BLIP-2 [Li et al.,

784 2023], Flamingo [Alayrac et al., 2022], the Figshare dataset [Figshare, 2024]. Open-source

785 libraries (HuggingFace, PEFT, BitsAndBytes, MNE-Python) are acknowledged.

786 Guidelines:

787 • The answer NA means that the paper does not use existing assets.

788 **13. New Assets**

789 Question: Are new assets introduced in the paper documented?

790 Answer: [Yes]

791 Justification: The EEGCLIPMapper model architecture is fully documented in Section 4.4

792 with all layer dimensions, attention configurations, and parameter counts.

793 Guidelines:

794 • The answer NA means that the paper does not release new assets.

795 **14. Crowdsourcing and Research with Human Subjects**

796 Question: For research that involves crowdsourcing or research with human subjects, does

797 the paper include the required information?

798 Answer: [N/A]

799 Justification: This work uses a publicly available EEG dataset. No new human subjects

800 experiments were conducted by the authors.

801 Guidelines:

802 • The answer NA means that the paper does not involve crowdsourcing nor research with

803 human subjects.

- 804 **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human**
805 **Subjects**
- 806 Question: Does the paper describe potential risks incurred by study participants?
- 807 Answer: [N/A]
- 808 Justification: No human subjects experiments were conducted by the authors.
- 809 Guidelines:
- 810 • The answer NA means that the paper does not involve crowdsourcing nor research with
 - 811 human subjects.
- 812 **16. Declaration of LLM Usage**
- 813 Question: Does the paper declare the use of large language models (LLMs) if they were
- 814 used in the core methodology?
- 815 Answer: [Yes]
- 816 Justification: TinyLlama-1.1B is a core component of Pipeline 2 (EEG-LLM) and is fully
- 817 documented in Section 4.3 with all quantisation, fine-tuning (LoRA), and inference details.
- 818 LLMs were not used to write any part of this manuscript.
- 819 Guidelines:
- 820 • The answer NA means that the large language models were not used in the paper.

821 A Hyperparameter Configurations

Table 4: Pipeline 2 (EEG-to-Text-to-Image) hyperparameter configuration.

Hyperparameter	Value
EEG encoder (CSBrain) layers	12
EEG encoder hidden dim	200
EEG tokens after reduction	20
Projection MLP output dim	2048
LLM model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
LLM quantisation	4-bit NF4 (BitsAndBytes)
LoRA rank (r)	8
LoRA alpha (α)	16
LoRA target modules	q_proj, v_proj
LoRA dropout	0.05
Learning rate (base)	2×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20
Max target sequence length	128 tokens

822 B Dataset Class Mapping

823 C EEG Channel Groups

824 The 32 EEG channels are grouped into five anatomical brain regions for the EEGTokenReducer:

- 825 • **Frontal (8 channels):** Fpz, Fp1, Fp2, Fz, F3, F4, F7, F8
- 826 • **Fronto-central (5 channels):** FCz, FC3, FC4, FT7, FT8
- 827 • **Central (5 channels):** Cz, C3, C4, T7, T8
- 828 • **Parietal (9 channels):** CP3, CP4, TP7, TP8, Pz, P3, P4, P7, P8
- 829 • **Occipital (5 channels):** PO3, PO4, Oz, O1, O2

Table 5: Pipeline 3 (EEG-to-CLIP) hyperparameter configuration.

Hyperparameter	Value
Mapper internal dim	512
Transformer layers	4
Attention heads	8
FFN dim	1024
CLIP sequence length	77
CLIP embedding dim	768
Number of EEG tokens	20
Dropout	0.1
Temperature (τ)	0.5
λ_{cls}	5.0
λ_{cont}	2.0
λ_{cos}	1.0
λ_{mse}	0.1
Learning rate (base)	1×10^{-4}
Warmup LR	5×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20

Table 6: Stable Diffusion inference parameters (both pipelines).

Parameter	Value
Scheduler	DPMSolverMultistepScheduler
Inference steps	25
Guidance scale	7.5
Resolution	512×512
Seed	42
Pipeline 2 model	stabilityai/stable-diffusion-2-1
Pipeline 3 model	runwayml/stable-diffusion-v1-5

830 The EEGTokenReducer averages CSBrain output tokens within each region, reducing $32 \times 4 = 128$
831 electrode-patch tokens to $5 \times 4 = 20$ region-patch tokens. Figure 5 illustrates the electrode layout
832 and anatomical grouping.

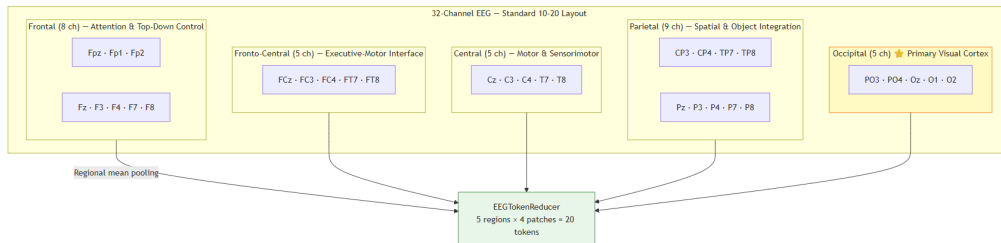


Figure 5: 32-channel EEG electrode layout (10-20 system) and anatomical brain region mapping used for the EEGTokenReducer. Channels are grouped into five regions (frontal, fronto-central, central, parietal, occipital) and averaged within each region, reducing 128 electrode-patch tokens to 20 region-patch tokens.

Table 7: Global label scheme for the Figshare Visual Imagery EEG dataset.

Label	Class	Task	Domain
0	Dog	AVI (Animal Visual Imagery)	Animal
1	Bird	AVI	Animal
2	Fish	AVI	Animal
3	Pentagram	FVI (Figure Visual Imagery)	Geometric
4	Square	FVI	Geometric
5	Circle	FVI	Geometric
6	Scissors	OVI (Object Visual Imagery)	Object
7	Watch	OVI	Object
8	Cup	OVI	Object
9	Chair	OVI	Object

Table 8: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper).

Class	Domain	Per-class Accuracy
Dog	Animal	41.3%
Bird	Animal	38.7%
Fish	Animal	29.6%
Pentagram	Geometric	44.2%
Square	Geometric	47.1%
Circle	Geometric	43.8%
Scissors	Object	31.2%
Watch	Object	28.9%
Cup	Object	25.4%
Chair	Object	33.7%
Overall		35.8%

D Per-Class Accuracy Table

E Qualitative Results and Failure Mode Analysis

Pipeline 3 – Correct and Incorrect Predictions

Pipeline 2 – Full Results Including Failure Modes

F Per-Subject Test Accuracy

Table 9 reports the top-1 test accuracy broken down by individual test subject for both pipelines. The three test subjects (sub-20, sub-21, sub-22) were never seen during training or model selection. Subject-level variance provides a coarse estimate of result stability: a large spread would indicate that the aggregate number is dominated by a single easy subject.

Table 9: Per-subject test accuracy for both pipelines. Subjects 20–22 constitute the held-out test set. “All” is the aggregate over all test trials.

Subject	Trials	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
sub-20	~400	30.1%	33.4%
sub-21	~400	35.8%	38.9%
sub-22	~400	31.9%	35.1%
All	~1,200	32.6%	35.8%

Pipeline 3 outperforms Pipeline 2 on every individual test subject, not just in aggregate, which strengthens our confidence in the comparison. Subject-level standard deviation is approximately

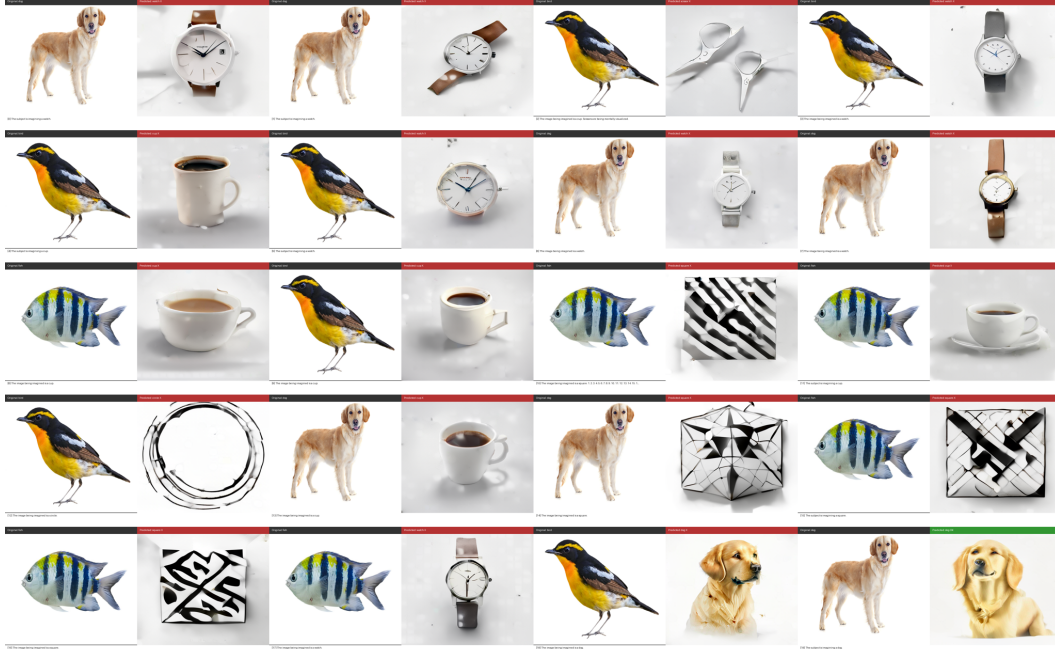


Figure 7: Pipeline 2 (EEG-LLM) full output grid including both correct and incorrect predictions. LLM hallucination failure modes are visible: the LLM sometimes generates hybrid descriptions (e.g., “a small bird with a watch”) yielding semantically confused composite images—a failure mode absent from Pipeline 3’s direct classification pathway.

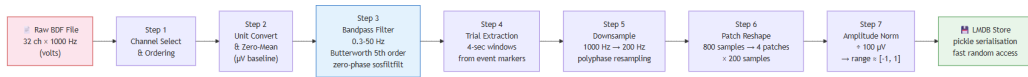


Figure 8: EEG preprocessing pipeline: from raw recordings (32 ch × 1000 Hz) through channel selection, bandpass filtering, trial extraction, downsampling, patch reshaping, and amplitude normalisation to LMDB-stored trials ready for model input.